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Variation in Form of the Pyramidal Tract and Its Relationship to Digital Dexterity[1]

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Abstract. A morphometric analysis of the pyramidal tract's relation to digital dexterity was performed on data from 69 mammals. The results show that the variation in digital dexterity among mammals corresponds most closely to the variation in place of termination of pyramidal tract fibers within the spinal cord, corresponds less closely to the variation in the size of the tract itself and its constituent fibers, and does not correspond reliably with any other feature yet reported. Since the termination of pyramidal tract fibers on or very near spinal motor neurons is a prerequisite even for the peculiar kind of dexterity seen in some non-primates (e.g., raccoon, kinkajou), this one feature alone seems to be a critical factor.

Introduction

On the basis of ablation studies on scattered mammalian species, there is a puzzling dichotomy in the function of the pyramidal tract between anthropoids and other mammals. In man or monkey, interruption of the pyramidal tract results in a weakness graded in increasing severity from proximal to distal and a loss of the ability to move the digits independently [TOWER, 1940; BUCY *et al.*, 1966; LAWRENCE and KUYPERS, 1968a, b]. In marked contrast, direct section of the pyramidal tract or ablation of its cortical origin in non-anthropoids has little apparent effect. In dogs, cats, prosimians, phalangers, or opossums, pyramidal tract

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destruction results in an initial weakness sometimes lasting several weeks, but it does not result in a paralysis reminiscent of that seen in anthropoids nor in any obvious or permanent deficit [BROMILEY and BROOKS, 1940; BUXTON and GOODMAN, 1967; CHAMBERS and LIU, 1957; HORE *et al.*, 1973; JANE *et al.*, 1965; REES and HORE, 1970].

The increase in anatomical information over the last 20 years [LASSEK, 1954] makes it possible to suggest that the marked differences in type or degree of deficit parallel the fine gradations in pyramidal tract anatomy. In general, the mammals which suffer the greatest deficit as a result of damage to their pyramidal tract are those with the most highly developed digital skills; and those which are most dexterous seem to be those with the most highly developed pyramidal tracts. It is this latter relationship between the variation in form of the pyramidal tract and the variation in degree of digital dexterity which is the primary focus of this report.

During the past half century, numerical data describing the pyramidal tract in a variety of mammals have become available. This pool of data has now grown to a point which allows a strict comparative inquiry using only commonplace statistical techniques. Accordingly, we have collected an array of from two to nine sets of descriptive data on pyramidal tract morphology and digital dexterity in a total of 69 mammalian species and subjected it to a statistical analysis which serves to identify several reliable relationships between digital dexterity and variations in the pyramidal tract or its sites of termination.

Materials and Methods

The data for this report were derived almost entirely from the published work of others. The reports contributing numerical data are listed in table I and designated in the bibliography by an asterisk. In general, three classes of descriptive parameters were specifically included in the analysis: (1) two parameters describing the animals themselves (body weight and body length); (2) four parameters describing the pyramidal tract and its constituent fibers (area of the tract, number of fibers per tract, largest fiber diameter, and average fiber size), and (3) three parameters describing the tract's terminations in the spinal cord (depth of penetration down the spinal cord, deepest, or ventralmost, lamina in the spinal gray with pyramidal tract terminations, and lamina in the spinal gray receiving the densest pyramidal tract terminations). The relationships between these parameters were examined first singly in both linear and logarithmic forms, and then in several combinations, for their power to account for digital dexterity.

Table I. Animals and anatomical data on which all correlations are based

Animal	Digital dexterity	Body weight kg	Number of fibers per tract	Largest fiber diameter μm	Average fiber size $\times 10^5$ K	Area of tract mm^2	Penetration down spinal cord	Lamina of axon terminals	
								deepest	densest
Aardvark *Orycteropus afer*	2	60				0.53 VERHAART, 1970	Upper C VERHAART, 1970		
Anteater *Myrmecophaga jubata*	1	20		10 VERHAART, 1970		0.60 VERHAART, 1970			
Armadillo *Dasypus novemcinctus*	1	5		2 VERHAART, 1970		0.60 VERHAART, 1970	T3 FISHER *et al.*, 1969	8 FISHER *et al.*, 1969	6
Bat (vampire) *Desmodus rufus*	1	0.033				0.01 YAMAMOTO *et al.*, 1955			
Bat (Indian tuber nose) *Harpyiocephalus leucogostra*	1	0.01				0.04 YAMAMOTO *et al.*, 1955			
Bat *Myotis*	1	0.03				0.01 VERHAART, 1970	S BROERE, 1966		
Bat (India Kalong) *Pteropus edulis*	1	0.9				0.12 VERHAART, 1970			

Table I (continued)

Animal	Digital dexterity	Body weight kg	Number of fibers per tract	Largest fiber diameter μm	Average fiber size $\times 10^5$ K	Area of tract mm^2	Penetration down spinal cord	Lamina of axon terminals	
								deepest	densest
Bat (horseshoe) *Rhinolophus*	1	0.016				0.01 YAMAMOTO *et al.*, 1955			
Bat *Tasarida cynocephala*	1	1.3	8,000		0.25	0.02 LASSEK and SHAPIRO, 1956			
Bear (assume Brown)	2	200				9.21 VERHAART, 1970			
Beaver *Castor canadensis*	3	13					S DOUGLAS and BARR, 1950		
Bushbaby *Galago crassicaudatus*	5	1.14					S FERRER, 1971	9 FERRER, 1971	6 FERRER, 1971
Bushbaby *Galago senegalensis*	5	0.26		2.5 VERHAART, 1966			Lower L VERHAART, 1966		
Camel *Camelus*	1	570		6 VERHAART, 1970					

Table 1 (continued)

Animal	Digital dexterity	Body weight kg	Number of fibers per tract	Largest fiber diameter μm	Average fiber size $\times 10^5$ K	Area of tract mm²	Penetration down spinal cord	Lamina of axon terminals	
								deepest	densest
Cat (domestic) *Felis catus*	2	3.5	186,000 LASSEK and RASMUSSEN, 1940	5 SCHEIBEL and SCHEIBEL, 1966	0.60	1.13 LASSEK and RASMUSSEN, 1940	S NYBERG-HANSEN and BRODAL, 1963	8 NYBERG-HANSEN and BRODAL, 1963	6 NYBERG-HANSEN and BRODAL, 1963
Chimpanzee *Pan troglodytes*	6	44	807,000 LASSEK and WHEATLEY, 1945		0.96	7.77 LASSEK and WHEATLEY, 1945	coccygeal PETRAS, 1968	10 PETRAS, 1968	6 PETRAS, 1968
Chipmunk *Tamias striatus*	3	0.1				0.26 SIMPSON, 1914	S SIMPSON, 1914		
Cow (unspecified)	1	500	540,000 LASSEK, 1942	5 VERHAART, 1970	0.55	2.97 LASSEK, 1942	Upper C LASSEK, 1942		
Coypu rat *Myocastor coypus*	3	5		8 GOLDBY and KACKER, 1963		0.37 GOLDBY and KACKER, 1963	S GOLDBY and KACKER, 1963	6 GOLDBY and KACKER, 1963	
Deer (unspecified)	1	200	490,000 LASSEK, 1942	6 VERHAART, 1970	0.58	2.89 LASSEK, 1942	Upper C LASSEK, 1942		

Table I (continued)

Animal	Digital dexterity	Body weight kg	Number of fibers per tract	Largest fiber diameter μm	Average fiber size $\times 10^5$ K	Area of tract mm²	Penetration down spinal cord	Lamina of axon terminals	
								deepest	densest
Dog (domestic) *Canis familiaris*	2	11	285,500 LASSEK and RASMUSSEN, 1940		1.22	3.49 LASSEK and RASMUSSEN, 1940	L5 BUXTON and GOODMAN, 1967	8 BUXTON and GOODMAN, 1967	
Elephant (Indian) *Elephas maximus*	1	6000		8 VERHAART, 1963		5.13 VERHAART, 1970	Middle T VERHAART, 1963		
Ferret *Putorius furo*	3	0.9	90,000 NOORDUYN, 1959	10 NOORDUYN, 1959		1.5 NOORDUYN, 1959			
Gibbon *Hylobates lar*	6	6	315,000 VERHAART, 1948b	12 VERHAART, 1970	1.09	3.44 VERHAART, 1970	S PETRAS, 1968	10 PETRAS, 1969	6 PETRAS, 1968
Gibbon *Hylobates syndactylus*	6	10	239,700 VERHAART, 1948b		1.35	3.25 VERHAART, 1948b			
Goat (domestic) *Capra hircus*	1	80	260,000 LASSEK, 1942	5 VERHAART, 1970	0.21	0.569 VERHAART, 1970	C7 HAARTSEN and VERHAART, 1967	6 HAARTSEN and VERHAART, 1967	

Table I (continued)

Animal	Digital dexterity	Body weight kg	Number of fibers per tract	Largest fiber diameter μm	Average fiber size $\times 10^5$ K	Area of tract mm²	Penetration down spinal cord	Lamina of axon terminals	
								deepest	densest
Gopher *Spermophilus tridecemlineatus*	3	0.65					L3 SIMPSON, 1915		
Guinea Pig *Cavia aperea*	2	0.6				0.33 REVELEY, 1915			
Hamster *Cricetus auratus*	3	0.5	480,000 DOUGLAS and BARR, 1950				S DOUGLAS and BARR, 1950		
Hedgehog *Erinaceous*	2	0.75				0.02 VERHAART, 1970	Upper C VERHAART, 1970		
Hedgehog *Paraechinus*	2	0.5					Upper C ROBARDS, personal commun.; CAMPBELL, 1967		
Hog (domestic) *Sus scrofa*	1	90	210,000 LASSEK, 1942	5 VERHAART, 1970	0.85	1.8 LASSEK, 1942			
Horse (domestic) *Equus caballus*	1	350		5 VERHAART, 1970		2.68 VERHAART, 1970			

Table 1 (continued)

Animal	Digital dexterity	Body weight kg	Number of fibers per tract	Largest fiber diameter μm	Average fiber size $\times 10^5$ K	Area of tract mm^2	Penetration down spinal cord	Lamina of axon terminals: deepest	Lamina of axon terminals: densest
Kinkajou *Potos flavus*	3	2.1					Lower L Harting and Noback, 1970	10 Harting and Noback, 1970	
Lemur *Lemur catta*	5	2.1		3 Verhaart, 1966			Lower L Verhaart, 1966		
Man *Homo sapiens*	7	70	1,101,000 Lassek and Rasmussen, 1940	22 Truex and Carpenter, 1969	1.03	11.43 Lassek and Rasmussen, 1940	coccygeal Truex and Carpenter, 1969		
Mole *Scalopus aquaticus*	1	0.1		4 Bishop *et al.*, 1971		0.01 Linowiecki, 1914	Upper T Linowiecki, 1914		
Monkey (spider) *Ateleus ater*	4	6	505,000 Lassek, 1943	12 Lassek, 1943	0.63	3.23 Lassek, 1943	coccygeal Petras, 1968	10 Petras, 1969	6 Petras, 1968
Monkey (capuchin) *Cebus apella*	5	1.25				2.0 Manocha *et al.*, 1968	coccygeal Petras, 1968	10 Petras, 1969	6 Petras, 1968
Monkey (vervet) *Cercopithecus pygerythrus*	6	4					coccygeal Petras, 1968	10 Petras, 1969	6 Petras, 1968

Table I (continued)

Animal	Digital dexterity	Body weight kg	Number of fibers per tract	Largest fiber diameter μm	Average fiber size $\times 10^5$ K	Area of tract mm^2	Penetration down spinal cord	Lamina of axon terminals	
								deepest	densest
Monkey (wooly) *Lagothrix lagotricha*	5	6					coccygeal PETRAS, 1968	10 PETRAS, 1968	6 PETRAS, 1968
Monkey (macaque) *Macaca ira*	6	5	195,000 VERHAART, 1948a		0.60	1.18 VERHAART, 1954			
Monkey (macaque) *Macaca mulatta*	6	8.3	400,000 DEMEYER and RUSSEL, 1958		0.72	2.89 LASSEK, 1941	coccygeal PETRAS, 1968	10 PETRAS, 1969	6 PETRAS, 1968
Monkey (baboon) *Papio papio*	6	25						10 PHILLIPS, 1967	
Monkey (marmoset) *Saquinas oedipus*	4	0.4					S SHRIVER and MATZKE, 1965	6 SHRIVER and MATZKE, 1965	
Monkey (squirrel) *Saimiri sciureus*	5	0.75		8 VERHAART, 1970		0.70 VERHAART, 1970	S VERHAART, 1966	9 HARTING and NOBACK, 1970	6 HARTING and NOBACK, 1970

Table I (continued)

Animal	Digital dexterity	Body weight kg	Number of fibers per tract	Largest fiber diameter μm	Average fiber size $\times 10^5$ K	Area of tract mm^2	Penetration down spinal cord	Lamina of axon terminals	
								deepest	densest
Mouse (unspecified)	3	0.025	32,300 LASSEK and RASMUSSEN, 1940		0.21	0.07 LASSEK and RASMUSSEN, 1940			
Mule *Equus*	1	250	412,000 LASSEK, 1942	5 VERHAART, 1970	0.90	3.74 LASSEK, 1942			
Opossum *Didelphis*	3	3.7	48,100 TOWE, 1973		0.45	0.22 TOWE, 1973	Upper T BAUTISTA and MATZKE, 1965	8 MARTIN and FISHER, 1968	5 MARTIN and FISHER, 1968
Orangutan *Pongo pygmaeus*	6	60				2.61 VERHAART, 1970			
Pangolin *Manis pentadactyla*	1	18		2.5 VERHAART, 1970		0.16 HSIANG-TUNG, 1944			
Phalanger *Trichosurus vulpecula*	3	3.5				0.54 MARTIN *et al.*, 1970	Lower T REES and HORE, 1970 MARTIN *et al.*, 1970	8 REES and HORE, 1970 MARTIN *et al.*, 1970	6 REES and HORE, 1970 MARTIN *et al.*, 1970

Table 1 (continued)

Animal	Digital dexterity	Body weight kg	Number of fibers per tract	Largest fiber diameter μm	Average fiber size $\times 10^5$ K	Area of tract mm^2	Penetration down spinal cord	Lamina of axon terminals	
								deepest	densest
Pygmy shrew *Microsorex hoyi*	2	0.003		1 VERHAART, 1970		0.02 VERHAART, 1970			
Quokka wallaby *Setonix brachyurus*	3	3.5				0.10 WATSON, 1971	T7 WATSON, 1971	6 WATSON, 1971	5 WATSON, 1971
Rabbit (domestic) *Oryctolagus cuniculus*	2	1.8	101,700 LASSEK and RASMUSSEN, 1940	4 VERHAART, 1970	0.27	0.28 LASSEK and RASMUSSEN, 1940	C4 VERHAART, 1970		
Raccoon *Procyon lotor*	3	12					S SIMPSON, 1912	10 PETRAS and LEHMAN, 1966	
Rat (unspecified)	3	0.4	137,000 BROWN, 1971	5 BISHOP *et al.*, 1971	0.10	0.14 BROWN, 1971	S GOODMAN *et al.*, 1966	8 GOODMAN *et al.*, 1966	5 BROWN, 1971
Rock hyrax *Procavia capensis*	2	17		9 VERHAART, 1967		0.72 VERHAART, 1970	S VERHAART, 1967		6 VERHAART, 1967

Table I (continued)

Animal	Digital dexterity	Body weight kg	Number of fibers per tract	Largest fiber diameter μm	Average fiber size $\times 10^5$ K	Area of tract mm^2	Penetration down spinal cord	Lamina of axon terminals	
								deepest	densest
Seal *Callorhinus ursinus*	1	52	748,000 LASSEK and KARLSBERG, 1956	25 LASSEK and KARLSBERG, 1956; VERHAART, 1970	0.27	2.08 VERHAART, 1970			
Sheep (domestic) *Ovis aries*	1	80	238,000 LASSEK, 1942	5 VERHAART, 1970	0.59	1.42 LASSEK, 1942	C1 KING, 1911		6 KING, 1911
Slow loris *Nycticebus coucang*	5	1.3		4.5 VERHAART, 1970		0.28 VERHAART, 1966	S CAMPBELL, 1967	10 CAMPBELL *et al.*, 1966	6 CAMPBELL, *et al.*, 1966
Spiney anteater *Tachyglossus aculeata*	2	4.2				0.39 VERHAART, 1970	T16 GOLDBY, 1939		
Squirrel *Sciurrus hudsonius*	3	0.6				0.40 SIMPSON, 1914	S SIMPSON, 1914		

Table I (continued)

Animal	Digital dexterity	Body weight kg	Number of fibers per tract	Largest fiber diameter μm	Average fiber size × 10⁵ K	Area of tract mm²	Penetration down spinal cord	Lamina of axon terminals	
								deepest	densest
Tapir (unspecified) *Tapirus*	1	250		5 VERHAART, 1970		1.48 VERHAART, 1970			
Tarsier *Tarsius*	4	0.12				0.16 TILNEY, 1927			
Tasmanian potoroo *Potorous apicalis*	2	1.35					T12 MARTIN *et al.*, 1972	7 MARTIN *et al.*, 1972	5 MARTIN *et al.*, 1972
Tree shrew *Tupaia glis*	3	0.16		2 VERHAART, 1966		0.12 TIGGES and SHANTHA, 1969	Lower T SHRIVER and NOBACK, 1967	6 JANE *et al.*, 1969	5 CAMPBELL, 1967
Whale (porpoise) *Phocaena phocoena*	1	60		5 VERHAART, 1970		1.27 VERHAART, 1970			
Woodchuck *Marmota monax*		5.3				0.37 DOUGLAS and BARR, 1950	S DOUGLAS and BARR, 1950		

Parameters Describing the Animal's Body

Since the size of an animal sets limits on the size of its neurological structures, it was important to determine the influence of body size, as measured by weight and length, on the value of the other parameters discussed in this report.

Body weight. The body weights assigned to each animal were taken from one of four sources: (1) the same report as the neuroanatomical data; (2) *Mammals of the World* [WALKER, 1968]; (3) *A Handbook of Living Primates* [NAPIER and NAPIER, 1967]; (4) TOWE [1973]. Whenever two of the secondary sources disagreed on Primates, the figures of NAPIER and NAPIER [1967] were selected. For domestic animals, the weights given by TOWE [1973] were used. In cases in which only a range of weights was given (including cases of marked sexual dimorphism) the midpoint of the range was used for statistical purposes.

Body length. As with weight, the body length for each animal was obtained either from the relevant neuroanatomical reports or from standard references [NAPIER and NAPIER, 1967; WALKER, 1968]. Since, as it turned out, this measure added nothing to the analysis beyond that contributed by body weight, no further mention of it seems warranted.

Parameters Describing the Pyramidal Tract

Tract size. The size of the pyramidal tract varies greatly among mammals. This variation has engendered considerable comment since the tract's discovery by PETIT in 1710. Recently, VERHAART [1970] has suggested that the size of the tract is a particularly important factor in determining its overall level of development.

As a measure of the size of the pyramidal tract, the area of a single pyramid at the level of the lower medulla has been used here. When the area of the tract at the lower medulla was not available, a reasonable estimate was made by one of several techniques. For example, if an author did not give the measurement but did provide a photomicrograph of the appropriate level and stated the magnification, we traced the border of the tract and then measured its area with a planimeter.

Nevertheless, several cases required special treatment: guinea pig, squirrel, and four bats. For guinea pig and squirrel an average of two measures had to be taken, the first from the upper medulla and the second from the upper cervical spinal cord. For two bats, *Harpyiocephalus* and *Pteropus,* the only measures available were from the lower pons and these were used despite being an obvious overestimate. For two other bats, *Desmodus* and *Rhinolophus,* no pyramidal tract has been reported even at the level of the upper pons. But since it was necessary for some potentially interesting comparisons to avoid absolute zero, we arbitrarily assigned a very small non-zero value: 0.01 mm^2. But because of the large number of mammals for which a reasonably accurate estimate of pyramidal tract size is available (48), these inaccuracies in the estimates for bats have very little statistical effect and no effect whatever on the main conclusions.

Number of fibers in the tract. The number of fibers in the pyramidal tract is known to vary widely among mammals but there has been disagreement as to the importance of this factor. LASSEK [1954], in his review, concluded that neither size nor skill of an animal was related to the number of fibers in its pyramidal tract. Apparently, it was this belief that prompted him to state that the function of the